

IndicIPR-QA: A Benchmark for Extractive Question Answering on Indian IPR Laws

1st First Author Name

Department Name

Institution Name

City, Country

email@domain.com

2nd Second Author Name

Department Name

Institution Name

City, Country

email@domain.com

Abstract—Legal Question Answering (QA) over statutory texts is a challenging task in Natural Language processing (NLP), especially in low-resource countries with complex legal frameworks. However, there is currently no extractive QA benchmark for Indian Intellectual Property Rights (IPR) laws, which contain specialized legal terms, nested sections, and long multi-clause answers across seven Acts. In this paper, we present IndicIPR-QA¹, an extractive QA benchmark dataset related to Indian IPR laws. The proposed dataset contains 4,462 QA pairs collected from 1,116 statutory paragraphs across seven Acts. To ensure both diversity and quality, the questions and answers are generated using GPT-4o through a structured prompting pipeline and subsequently validated using nine automatic quality filters, an LLM-as-Judge single and ensemble evaluation, and a human validation study ($\kappa_{\text{QW}} = 0.328$, 95.0% answer correctness). Experiments show that BM25 achieves a token-level F1 score of only 20.21%, whereas a fine-tuned InLegalBERT model attains an F1 score of 80.56%. The substantial performance gap of 60.35 points highlights the challenging nature of the benchmark and suggests that effective performance cannot be achieved by simple lexical matching alone.

Index Terms—IndicIPR-QA Dataset, Legal NLP, Extractive QA, Indian IPR Law, Quality Validation.

I. INTRODUCTION

Intellectual Property Rights (IPRs) protect the creations of the human intellect including inventions, trademarks, copyright works, designs, geographical indications, semiconductor layout designs, and plant varieties. Indian IPR law is written across seven major Acts – the Patents Act (1970), Trade Marks Act (1999), Copyright Act (1957), Designs Act (2000), Geographical Indications of Goods Act (1999), Semiconductor Integrated Circuits Layout-Design Act (2000), and Protection of Plant Varieties and Farmers’ Rights Act (2001). Every Act contains its own terminology, guidelines, processes, exceptions, and cross-references. Consequently, extracting the precise legal solution from these Acts can be a lengthy process. Accessing legal information is challenging due to the length and complexity of statutory documents. Although laws already contain answers to issues such as trademarks, patents, design protection, and copyright remedies, locating the exact provisions is often time-consuming even for legal professionals.

Automated question answering (QA) systems can help reduce this problem. These systems can take a natural-language question and return the answer directly from the legal text. However, a major data gap remains. Existing QA datasets are mostly general-domain (SQuAD [1], TriviaQA), common-law based (CUAD [2]), non-extractive (JEC-QA [3]), or focused on court judgments rather than statutory Acts. As a result, no existing dataset directly supports extractive QA over Indian statutory IPR documents. This paper adopts the extractive QA approach, where the answer is an exact span from the source text. Such approach is suitable for law domain because legal answers must be precise and easy to verify. In summary, the major contributions of this paper can be outlined as follows:

- 1) It presents the generation of IndicIPR-QA, the first extractive question answering dataset covering all seven major Indian IPR Acts. It contains 4,462 question-answer pairs from 1,116 unique statutory paragraphs. The dataset is split at the section level to ensure there is no context leakage (Section III).
- 2) The quality of the dataset is validated through a nine-step quality-control pipeline, an LLM-as-Judge ensemble, and a human validation study with two law graduates on a 200-pair stratified sample ($\kappa_{\text{QW}} = 0.328$, 95.0% answer correctness) (Section III).
- 3) It provides a linguistic characterization and difficulty analysis. IndicIPR-QA has the longest average answer length (35.6 words) among comparable extractive legal QA datasets. The analysis also shows that question-context lexical overlap ($r = -0.864$) is the strongest predictor of per-Act difficulty (Section IV).
- 4) Four BERT family encoders and three open LLMs are benchmarked. The 60.35 point F1 gap between BM25 (20.21) and InLegalBERT (80.56) indicates that IndicIPR-QA requires contextual understanding beyond keyword matching (Section V).

II. RELATED WORK

Existing research shows that several benchmark datasets have been proposed for question answering and reasoning over legal texts, covering tasks such as extractive QA, statutory retrieval, legal reasoning, contract understanding, and judgment analysis.

¹Dataset: <https://huggingface.co/datasets/niloycste68/IndicIPRExtractiveQA>;
Code: <https://github.com/niloycste/Indic-IPR-Dataset-.git>

In 2016, Rajpurkar et al. established the standard extractive QA setting SQuAD, where answers are selected as spans from a given passage [1]. SQuAD 2.0 extended this task by adding unanswerable questions. Although these datasets are widely used, they are based on general-domain text and do not capture the complexity of legal statutes.

Several legal QA and retrieval datasets have been developed in recent years. CUAD [2] provides expert-annotated QA pairs from commercial contracts, while LegalBench evaluates legal reasoning across different task formats. JEC-QA [3] focuses on multiple-choice legal questions from the Chinese judicial examination. COLIEE, BSARD, and ALQAC support statutory article retrieval, wherein systems identify relevant legal provisions for a query. However, many of these datasets focus on retrieval, contracts, examination-style questions, or non-Indian legal systems rather than extractive answer span prediction from Indian statutes.

Recent datasets such as PrivacyQA [4], LLeQA [5], and VLQA [6] address legal question answering using privacy policies, legal consultation data, and real-world citizen questions. These works show the importance of building legal QA systems for practical public use. However, long-form generation may produce fluent yet unsupported answers, which makes extractive QA important for legal domains where answers must remain grounded in the source text.

Indian legal NLP research has focused mainly on judgment prediction, judgment summarization, rhetorical role labeling, and legal language models [7]. Recent works such as AILQA [8] explore QA over Indian legal documents, but they do not provide an extractive benchmark grounded in Indian IPR statutes. LLM-generated datasets have also become useful for scalable data construction, but they require careful filtering and validation to reduce hallucination and noisy annotations.

Most of the existing datasets are either designed for general-domain applications, focused on foreign legal systems such as those of the United States or Europe, or primarily based on court judgments and case law rather than statutory law. Overall, the existing works lack an extractive QA benchmark for Indian statutory law, especially in the specialized field of intellectual property rights. To address this gap, this work proposes IndicIPR-QA, an extractive question-answering benchmark over Indian IPR statutes.

III. DATASET CONSTRUCTION

The dataset construction proceeded in five phases: corpus creation, QA generation, quality control, section-level splitting, and audit/validation. Fig. 1 and Algorithm 1 provide an overview of the full pipeline – from raw statute text to the curated benchmark.

A. Source Corpus and QA Generation

The full text of all seven Indian IPR Acts has been extracted from official Government of India PDFs on the India Code portal (<https://www.indiacode.nic.in/>) using pdfplumber. During preprocessing, table of contents pages are removed, editorial amendment markers (such as “Ins.,” “Subs.,” “w.e.f.”) are

Algorithm 1 IndicIPR-QA: From raw statutes to curated benchmark

Input: Act PDF set $\mathcal{P} = \{P_1, \dots, P_7\}$; chunk size $\delta = 1400$
Output: Curated dataset $\mathcal{D}$, retained contexts $\mathcal{C}^*$, and splits $(\mathcal{D}_{tr}, \mathcal{D}_{va}, \mathcal{D}_{te})$

Step 1: Corpus extraction

```

1:  $\mathcal{C} \leftarrow \emptyset$ 
2: for all  $P_i \in \mathcal{P}$  do
3:    $T_i \leftarrow \text{EXTRACTTEXT}(P_i)$  // TOC removal, body start detection
4:    $T_i \leftarrow \text{CLEANTEXT}(T_i)$  // Remove editorial lines, normalize whitespace
5:    $\mathcal{C} \leftarrow \mathcal{C} \cup \text{CHUNKCONTEXTS}(T_i, \delta)$  // Sentence-safe chunking
6: end for
7:  $\mathcal{C} \leftarrow \text{VALIDATECONTEXTS}(\mathcal{C})$  // 1,173 valid contexts

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Step 2: QA pair generation

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8:  $\mathcal{R} \leftarrow \emptyset$ 
9: for all  $c \in \mathcal{C}$  do
10:   $\mathcal{R} \leftarrow \mathcal{R} \cup \text{GENERATEQA}(c)$  // GPT-4o; up to 5 extractive QA pairs
11: end for

```

Step 3: Quality control and context-balanced selection

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12:  $\mathcal{R} \leftarrow \text{APPLYFILTERS}(\mathcal{R}, F_1, \dots, F_9)$  // 4,689 pairs after automatic filtering
13:  $\mathcal{D} \leftarrow \text{CONTEXTBALANCEDSELECT}(\mathcal{R}, 4462)$  // Score-based selection  $\rightarrow$  4,462 pairs
14:  $\mathcal{C}^* \leftarrow \text{RETAINCOVEREDCONTEXTS}(\mathcal{C}, \mathcal{D})$  // 1,116 retained contexts

```

Step 4: Section-level dataset splitting

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15:  $\mathcal{G} \leftarrow \text{GROUPBYSECTION}(\mathcal{D})$  // 681 unique statute sections
16:  $(\mathcal{D}_{tr}, \mathcal{D}_{va}, \mathcal{D}_{te}) \leftarrow \text{STRATIFIEDSPLIT}(\mathcal{G}, 80/10/10)$  // Per-Act split, seed 42

```

Step 5: Audit and validation

```

17:  $\text{LLMJUDGEAUDIT}(\mathcal{D}, \text{GEMMA-3:12B})$  // Single-judge audit
18:  $\text{LLMJUDGEAUDIT}(\mathcal{D}, \text{ENSEMBLE-3})$  // Three-model ensemble
19:  $\text{HUMANVALIDATION}(\mathcal{D})$  // 200-pair sample, 2 annotators
20: return  $\mathcal{D}, \mathcal{C}^*, (\mathcal{D}_{tr}, \mathcal{D}_{va}, \mathcal{D}_{te})$ 

```

stripped, and whitespace is normalized. The cleaned text is then split into 1,173 paragraph-level contexts, each around 1,400 characters while keeping sentences and section boundaries intact.

For each context, GPT-4o [9] is used to generate up to five extractive QA pairs. GPT-4o is used as it follows instructions well and reproduces statutory text verbatim without paraphrasing. The prompt requires answers to be exact substrings of the context, questions to target specific legal elements, and vague section summaries or yes/no questions to be avoided. This process produces 5,721 raw candidate pairs.

B. Nine-Step Quality Control Pipeline

All 5,721 raw QA pairs passed through nine sequential automatic filters (F1 to F9). F1 checks for non-empty fields while F2 ensures that the answer appears as an exact verbatim span in the context, removing hallucinated answers. F3 keeps answer lengths between 5 and 350 characters; F4 removes answers covering more than 85% of the context; F5 checks question quality (at least 5 words, ends with “?”, and avoids vague references); F6 removes questions that directly contain the answer. Whereas, F7 removes truncated contexts that begin or end incompletely and F8 removes editorial amendment markers from answers. Finally, F9 removes answers that are only standalone 1–4 digit numbers.

Among all nine filters F3 and F4 are the most influential filtering stages, removing 595 and 417 pairs respectively, accounting for 1,012 from the 1,032 automatically rejected pairs. Their removal indicates that GPT-4o predominantly generated answers that are either uninformatively short or excessively long, often reproducing large portions of the source section. After automated filtering, 4,689 QA pairs satisfied all nine quality checks. A context-balanced selection step then reduced the dataset to 4,462 QA pairs from 1,116 unique

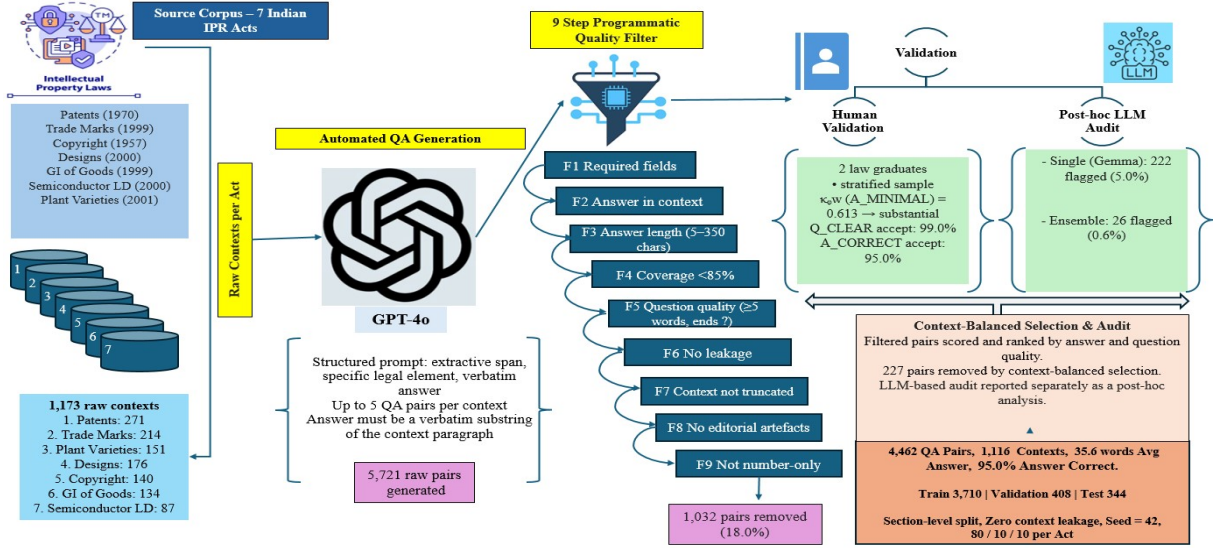


Fig. 1. End-to-end IndicIPR-QA construction: QA generation, nine-step filtering, context-balanced selection, and post-hoc LLM audit.

contexts. This step groups pairs by context, scores each pair by answer and question length, and algorithmically selects ranked pairs to reach the target size while ensuring balanced coverage across all contexts. The importance of this selection step is confirmed empirically: omitting it reduces test F1 by 3.05 points (Table II), the largest single degradation among all pipeline variants tested. Then, the dataset is partitioned into train, validation, and test splits (3,710/408/344) at the section level to ensure zero contextual overlap across splits.

C. Pipeline Ablation Study

To assess the contribution of each filter, in this work an ablation study has been performed where one filter is disabled at a time while the remaining eight are kept active. Table I report the number of additional potentially noisy pairs that would have entered the automatically accepted set if each filter were disabled.

The experiment results show that F3 and F4 are the most influential filtering steps. Disabling F3 admits 368 additional potentially noisy pairs (+7.8% relative to the 4,689-pair baseline), while disabling F4 admits 417 additional pairs (+8.9%). These results indicate that the primary GPT-4o failure mode is incorrect answer span length rather than hallucination: generated answers are often either too short to convey meaningful legal information or excessively long spans that reproduce large portions of the statutory text.

D. Model-Level Filter Ablation

To verify that the quality of this pipeline improves real model performance and not just dataset cleanliness, InLegalBERT is trained on six different training data variants while keeping the validation and test sets fixed to the same clean split.

Table II shows that every relaxation of the quality pipeline degrades test F1. The largest observed effect is from context-balanced selection: removing it (heuristic all) costs 3.05 F1 points, the largest single drop, suggesting that the final selection

TABLE I
QUALITY-FILTER ANALYSIS OF THE 5,721 RAW GPT-4O GENERATED CANDIDATE QA PAIRS. SEQUENTIAL FILTERING APPLIES FILTERS F1–F9 IN ORDER. PER-FILTER ABLATION OMITTS EACH FILTER INDIVIDUALLY WHILE KEEPING THE REMAINING FILTERS ACTIVE.

ID	Filter criterion	Sequential filtering		Per-filter ablation	
		Removed	Retained	Recovered pairs	Accepted
F1	Required fields	0	5,721	0 (0.0%)	4,689
F2	Answer in context	0	5,721	0 (0.0%)	4,689
F3	Answer length 5–350 [†]	595	5,126	368 (7.8%)	5,057
F4	Coverage <85% [‡]	417	4,709	417 (8.9%)	5,106
F5	Question quality	13	4,696	13 (0.3%)	4,702
F6	No answer leakage	1	4,695	1 (0.0%)	4,690
F7	Context not truncated	3	4,692	3 (0.1%)	4,692
F8	No editorial artefacts	3	4,689	3 (0.1%)	4,692
F9	Answer not num-only	0	4,689	0 (0.0%)	4,689
		1,032	4,689		

[†]F3 removes 595 pairs during sequential filtering. When F3 is omitted, only 368 pairs are recovered because the remaining 227 F3-rejected candidates are still removed by one or more of the other active filters.

[‡]F4 removes 417 pairs during sequential filtering. When F4 is omitted, all 417 pairs are recovered, increasing the accepted set from 4,689 to 5,106.

improves training quality beyond what automated filters alone achieve. F4 (coverage <85%) contributes 1.84 F1 points when removed in isolation: without it, near-complete section copies enter training and teach the model to reproduce whole paragraphs rather than precise spans. F3 (answer length) has a smaller individual effect (−0.60), but its interaction with F4 is non-additive: removing both together costs only −0.20, suggesting their retained pairs partially overlap in failure mode.

E. LLM-as-Judge Quality Verification

To obtain an additional automated assessment of the 4,462 retained QA pairs beyond the rule-based filtering, an LLM-as-Judge protocol is employed under two different configurations and their outputs are compared. These scores are used only

TABLE II

MODEL-LEVEL FILTER ABLATION USING INLEGALBERT ON DIFFERENT TRAINING-DATA VARIANTS. $\Delta F1$ IS MEASURED RELATIVE TO THE CLEAN SELECTED BASELINE.

Training variant	F1	Char-F1	$\Delta F1$
clean selected (ours)	79.42	83.74	—
without F3+F4 combined	79.22	84.00	−0.20
without F3 (answer length)	78.82	83.61	−0.60
raw candidates (unfiltered)	77.59	81.70	−1.83
without F4 (coverage)	77.58	82.07	−1.84
heuristic all (no selection)	76.37	82.39	−3.05

TABLE III

LLM-AS-JUDGE EVALUATION WITH A SINGLE JUDGE AND A THREE-MODEL ENSEMBLE. SCORES RANGE FROM 1 (POOR/FLAGGED) TO 3 (GOOD).

Criterion	Single Judge		Ensemble	
	Avg. score	Flagged	Avg. score	Flagged
Question specificity	2.92	1.1%	2.93	0.0%
Answer sufficiency	2.92	2.8%	2.99	0.4%
Answer minimality	2.52	4.2%	2.85	0.4%
Legal relevance	2.97	0.7%	3.00	0.0%
Flagged QA pairs	222 pairs (5.0%)		26 pairs (0.6%)	

as auxiliary diagnostic indicators and not as a replacement for human evaluation.

Gemma 3:12b [10] is chosen for its strong instruction following capability at the 12B scale and fully open weights, ensuring reproducible judging without API rate limits. It scores all pairs at temperature 0 on four criteria: Q_SPECIFIC (the question targets a specific legal element), A_SUFFICIENT (the answer fully answers the question), A_MINIMAL (the answer is the shortest correct span), and LEGAL_RELEVANT (the question concerns substantive legal content). Scores are 1 (poor), 2 (acceptable), or 3 (good); a pair is flagged if any criterion scores 1.

Gemma 3:12b, Llama 3.1:8b, and Mistral:7b score the same four criteria independently, with majority vote determining the final score. Table III reports both settings.

The largest difference between the two settings appears on A_MINIMAL, where the single judge flags substantially more pairs, while the ensemble is more permissive toward longer statutory answer spans. Manual inspection of all 26 ensemble flagged pairs showed that they are legally valid but verbose definitions rather than genuine generation errors. Overall, the LLM-as-Judge analysis is useful for identifying potential failure patterns and score distributions, while the human validation study provides the stronger evidence of dataset quality.

F. Human Validation Study

Two law graduates independently annotate a stratified random sample of 200 QA pairs drawn from the test split (seed 42, all seven Acts: copyright 25, designs 24, GI 32, patents 29, plant varieties 28, semiconductor 30, and trade marks 32). At $n = 200$, the margin of error for an acceptance rate near 0.95 is $\pm 3.0\%$ at 95% confidence, sufficient to detect quality differences of practical significance. Each pair is rated on three ordinal criteria: Q_CLEAR (question clarity), A_CORRECT (answer correctness), and A_MINIMAL (answer minimality), using a 1–3 Likert scale.

TABLE IV

HUMAN INTER-ANNOTATOR AGREEMENT ON 200 QA PAIRS (TWO LAW GRADUATES). κ_{UW} = UNWEIGHTED COHEN’S κ ; κ_{QW} = QUADRATIC-WEIGHTED COHEN’S κ .

Criterion	κ_{UW}	κ_{QW}	Strength	Accept. %
Question Clarity	−0.008	0.051	slight [†]	99.0%
Answer Correctness	0.117	0.328	fair	95.0%
Answer Minimality	0.312	0.613	substantial	75.5% [‡]
Overall average	0.140	0.331	fair	—

[†] The near-zero κ for Q_CLEAR is caused by a highly skewed label distribution: 199 of 200 pairs received scores ≥ 2 from both annotators (99.0%). Under such ceiling conditions, chance-adjusted agreement metrics become unstable and are not meaningful.

[‡] Strict acceptability requires both annotators to assign a score ≥ 2 on A_MINIMAL. Under a lenient criterion, where at least one annotator assigns a score ≥ 2 , the acceptance rate increases to 82.0%. A total of 36 pairs were unanimously rejected by both annotators (score = 1).

For two annotators, inter-annotator agreement was measured using Cohen’s chance adjusted κ [11]:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (1)$$

where p_o is observed agreement and p_e is chance agreement. Because the labels are ordinal, we use quadratic weighted Cohen’s κ [12]:

$$\kappa_w = 1 - \frac{\sum_{i=1}^K \sum_{j=1}^K w_{ij} O_{ij}}{\sum_{i=1}^K \sum_{j=1}^K w_{ij} E_{ij}} \quad (2)$$

$$w_{ij} = \left(\frac{i - j}{K - 1} \right)^2 \quad (3)$$

The human validation results (Table IV) indicate strong overall dataset quality. Specifically, 99.0% of questions met the clarity criterion according to both the annotators, and 95.0% of answers are rated correct. Agreement on the answer minimality has reached the substantial range ($\kappa_{QW} = 0.613$), while answer correctness showed fair agreement ($\kappa_{QW} = 0.328$). The lower acceptance rate for minimality (75.5%) reflects the inherent subjectivity of determining concise answer spans in multi-clause statutory text. Figure 2 illustrates the agreement profile in all evaluation criteria.

IV. DATASET ANALYSIS

IndicIPR-QA is analyzed from both corpus level and Act specific perspectives to characterize its properties and complexity. Firstly descriptive statistics and comparison of the dataset with widely used QA benchmarks is given. Then, examine variations across individual intellectual property acts, focusing on lexical characteristics, answer complexity, and model performance. These analyses provide a comprehensive view of the datasets structure and the factors contributing to task difficulty.

TABLE V
COMPARISON OF INDICIPR-QA WITH RELATED QA BENCHMARKS (SQ1 = SQUAD 1.0; SQ2 = SQUAD 2.0; IPB = IPBENCH). “—” INDICATES UNAVAILABLE STATISTICS.

Property	SQ1	SQ2	CUAD	JEC	IPB	IndicIPR-QA(Ours)
QA pairs	87.6k	150k	13.1k	26.4k	10.4k	4.5k
Contexts	18.9k	35k	510	—	—	1,116
QA/context	4.8	4.3	25.7	—	—	4.0
Ans len (wds)	3.2	3.2	30.0	MCQ	MCQ	35.6
Ctx len (wds)	117	117	200	95	200	171.3
Q len (wds)	10.1	10.1	9.5	57	—	11.9
Human ann.	Yes	Yes	Yes	Yes	Yes	Yes[†]
Extractive	Yes	Yes	Yes	No	No	Yes
Unanswerable	No	Yes	Yes	No	No	No
Domain	General	General	Contracts	Chinese Bar Exam	US/Chinese IP Law	Indian IPR

[†]Validated on a stratified 200-pair sample with two law graduates ($\kappa_{QW} = 0.328$, 95.0% answer correctness).

A. Corpus Level Statistics

Table V places IndicIPR-QA in context with well-known QA benchmarks. IndicIPR-QA has the longest average answer length (35.6 words) among all extractive QA datasets in the comparison, which is $11 \times$ longer than SQuAD (3.2 words) and 19% longer than CUAD (30.0 words), making extractive span prediction substantially harder. The dataset is fully extractive with zero unanswerable questions and exhibits the strictest train–test separation by construction.

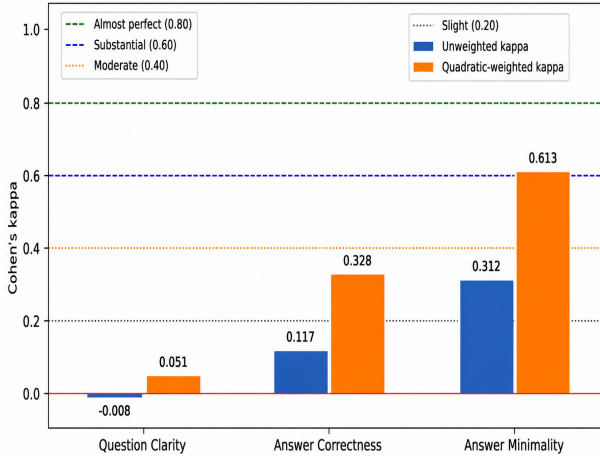


Fig. 2. Criterion-wise inter-annotator agreement for the human validation study. Answer minimality has the strongest agreement, while question clarity shows a ceiling effect due to skewed marginal distributions.

B. Per-Act Distribution and Difficulty

Table VI reports per-Act statistics together with test-set F1 scores for the strongest direct baseline, InLegalBERT. Performance varies substantially across Acts, ranging from 99.62% for the Designs Act to 52.80% for the Plant Varieties Act. Question–context lexical overlap is the strongest predictor of per-Act difficulty ($r = -0.864$, Pearson correlation). Interestingly, the Plant Varieties Act exhibits the highest lexical overlap (0.636) but the lowest F1 score, likely because its specialized biological terminology (e.g., “essentially derived variety” and “propagating material”) is poorly represented in the pretraining corpora of legal BERT models, causing models

TABLE VI
PER-ACT INDICIPR-QA STATISTICS, SORTED BY TEST F1. Q-CTX DENOTES STOPWORD-REMOVED QUESTION-CONTEXT UNIGRAM OVERLAP.

Act	n	Avg ans (words)	Q-Ctx ovlp	Test F1
Designs	173	43.4	0.320	99.62
Patents	1,143	35.8	0.429	96.53
Semicond.	585	22.1	0.418	86.86
GI	551	51.8	0.458	83.11
Copyright	349	62.3	0.447	78.03
Trade Marks	1,106	31.9	0.416	71.33
Plant Var.	555	21.9	0.636	52.80

to rely on misleading surface-level cues during span selection. In contrast, the Copyright Act contains a high proportion of multi-clause answers (53.9%), which correlates with lower performance (78.03% F1) compared with the Patents Act (13.7% multi-clause answers; 96.53% F1).

Lexical analysis shows that answers span several semantic categories, including other (31.4%), authority (29.4%), procedure (12.0%), rights (9.5%), conditions (7.9%), definitions (5.8%), duration (2.3%), and penalties (1.7%). The mean question-context unigram overlap, computed after removing stopwords, is 0.448. This indicates that fewer than half of the question words appear directly in the context, suggesting that the task requires more than simple keyword matching. In addition, 19.9% of answers contain semicolons, and 5.9% include the phrase “subject to”, reflecting the frequent presence of long and multi-clause statutory spans.

V. BASELINE EXPERIMENTS & RESULTS

Four BERT-based encoder models are fine-tuned on the IndicIPR-QA training split (3,710 QA pairs) using the Bert-ForQuestionAnswering architecture [13], which predicts the start and end positions of the answer span in the input sequence: [CLS] question [SEP] context [SEP]. The evaluated models represent different levels of domain specialization: BERT-base uncased [13] (general English), LegalBERT-US [14] (US legal text), InCaseLawBERT [15] (Indian court judgments), and InLegalBERT [7] (Indian legal text).

All encoder models use the same hyperparameters: the AdamW optimizer, learning rate 3×10^{-5} with 50 warm-up steps and linear decay, weight decay 0.01, effective batch size 32, maximum sequence length 384, document stride 128, 3

TABLE VII
BASELINE PERFORMANCE ON THE INDICIPR-QA TEST SPLIT (344 QA PAIRS). ZS = ZERO-SHOT; FS = 3-SHOT FEW-SHOT.

System	EM	F1	Char-F1
<i>Lexical retrieval</i>			
BM25 retrieval	0.00	20.21	77.10
BM25 oracle window	14.24	51.30	85.23
<i>Fine-tuned encoders</i>			
BERT-base	46.22	76.56	81.56
LegalBERT-US	47.38	79.75	84.02
InCaseLawBERT	44.77	77.88	82.50
InLegalBERT	46.80	80.56	84.92
<i>Prompted open LLMs</i>			
Mistral 7B (ZS)	18.31	67.11	75.07
Mistral 7B (FS)	24.71	68.83	76.01
Llama 3.1:8b (ZS)	33.43	74.78	79.61
Llama 3.1:8b (FS)	33.43	77.02	79.78
Qwen2.5:7b (ZS)	29.94	77.59	81.04
Qwen2.5:7b (FS)	34.30	76.39	78.60

training epochs, and random seed 42. These settings follow standard extractive QA fine-tuning practice [13]. The best checkpoint is selected using validation-set F1. This study also evaluates three open instruction-tuned LLMs in zero-shot and 3-shot settings: Llama 3.1:8b [16], Mistral 7B [17], and Qwen2.5:7b [18]. These models represent different model families at a similar 7–8B parameter scale. In zero-shot mode, the model receives the statute passage and question with instructions to copy the answer verbatim from the context. In 3-shot mode, three example QA pairs from the training set are provided as demonstrations.

As a lexical lower bound, BM25 [19] retrieval is evaluated. For a document D and query Q , the relevance score is

$$\text{BM25}(D, Q) = \sum_{q \in Q} \text{IDF}(q) \frac{f(q, D)(k_1 + 1)}{f(q, D) + k_1 \left(1 - b + b \frac{|D|}{\bar{D}}\right)} \quad (4)$$

where $f(q, D)$ is the term frequency of token q in document D , $|D|$ is the document length, $\bar{D}$ is the collection-average document length, and we use standard defaults $k_1 = 1.5$, $b = 0.75$.

A. Results

Table VII and Figure 3 summarize the results of all baseline systems. BM25 retrieval [19] achieves only 20.21 token-level F1, performing much worse than all supervised encoder models. Among the encoder models, InLegalBERT achieves the best F1 score (80.56), followed closely by LegalBERT-US (79.75). General-domain BERT-base obtains the lowest encoder performance (76.56), showing that domain-specific pretraining on legal text improves performance on statutory QA.

A paired bootstrap significance test with 10,000 resamples and Holm correction shows that InLegalBERT significantly outperforms BERT-base ($\Delta\text{F1} = +4.00$, $p = 0.002$) and InCaseLawBERT ($\Delta\text{F1} = +2.68$, $p = 0.036$). However, the performance difference between InLegalBERT and LegalBERT-US is not statistically significant ($\Delta\text{F1} = +0.81$, $p = 0.410$).

Among the prompted LLMs, Qwen2.5:7b achieves the best zero-shot performance ($\text{F1}=77.59$), while Llama 3.1:8b achieves the best 3-shot performance ($\text{F1}=77.02$). Even the strongest prompted LLM remains 2.97 F1 points below fine-tuned

InLegalBERT, indicating that supervised training on IndicIPR-QA provides a clear advantage for extracting long statutory answer spans. A key concern with automatically generated datasets is whether models rely on surface patterns rather than genuine language understanding. This issue has been examined in Section VI through zero-context leakage evaluation, novel answer vocabulary analysis, and the BM25-to-InLegalBERT performance gap.

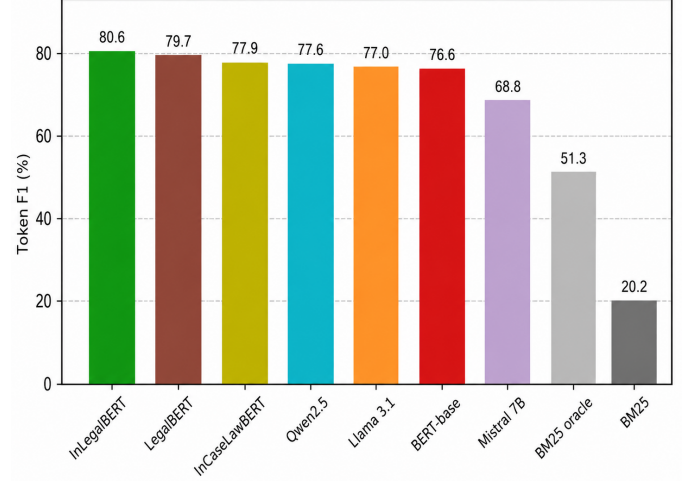


Fig. 3. Token-level F1 across all baselines on the IndicIPR-QA test split. The large gap between BM25 and InLegalBERT shows that lexical matching alone is insufficient for the benchmark.

VI. GENERALIZABILITY ANALYSIS

In the zero-context leakage setting, the dataset is split at the section level so that no paragraph context appears in both the training and test sets. The Jaccard similarity between train and test contexts is 0.00 for all seven Acts, and the novel test-context rate is 100%. (Figure 4)

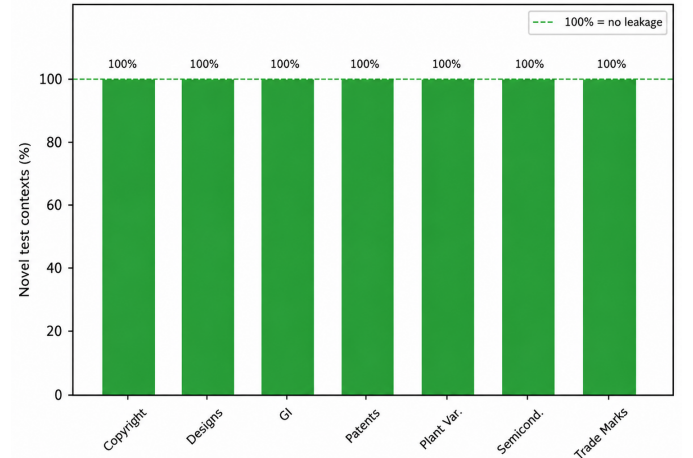


Fig. 4. Generalizability analysis of IndicIPR-QA: The 100% novel test-context rate across all Acts confirms zero train-test context overlap.

Regarding novel answer vocabulary, among the 1,234 unique answer tokens in the test set, 137 (11.1%) do not appear in

any training answer. In addition, 15.3% of test-set question tokens and 13.7% of test-set context tokens do not appear in the training split. This means that models must understand new legal terms instead of simply memorizing words from the training data. When comparing BM25 with learned models, BM25 retrieval achieves only 20.21 F1, while InLegalBERT reaches 80.56 F1, creating a large performance gap of 60.35 points. This suggests that the dataset cannot be solved through simple lexical matching alone. Even the BM25 oracle window, which is given the correct context, achieves only 51.30 F1, showing that extracting answer spans from long statutory passages remains difficult even when retrieval is perfect. (Figure 3)

VII. CONCLUSION

IndicIPR-QA is the first extractive QA benchmark that comprehensively covers all seven major Indian IPR Acts. The dataset contains 4,462 QA pairs generated using the GPT-4o API, refined through a nine-step quality-control pipeline and further validated via LLM- and human-based evaluation, achieving 95.0% answer correctness. It has the longest average answer length (35.6 words) among comparable extractive legal QA benchmarks, zero train-test context overlap, and 11.1% novel answer vocabulary in the test set. Finally, the large performance gap between BM25 retrieval (F1=20.21) and InLegalBERT (F1=80.56) shows that the benchmark requires contextual understanding rather than simple lexical matching. The per-Act results further show that lexical overlap alone does not guarantee strong performance; although, the Plant Varieties Act has the highest question-context lexical overlap (0.636), it achieves the lowest F1 score (52.80%), likely due to specialized biological and agricultural legal terminology and multiple similar candidate spans. This suggests that domain-adaptive pretraining on agricultural and biological legal text may improve performance. Future extensions may include scenario-based multilingual versions of the dataset covering additional Indian languages.

VIII. DATA AND CODE AVAILABILITY

The IndicIPR-QA dataset is publicly available on Hugging Face. The preprocessing pipeline, along with the training and evaluation scripts, is maintained in a GitHub repository and will be made publicly available upon acceptance of the paper.

- **Dataset:** IndicIPR-QA Dataset
- **Code:** Project Repository (to be released upon acceptance of the paper).

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